



AMET
UNIVERSITY

(Deemed to be University Under Section 3 of UGC Act 1956)

AMET-PHYSICS HORIZONS



Academic Year: 2024-25

EVEN Semester Magazine

AMET Deemed to be University, Chennai – 603 112

EDITORIAL

Dear Readers

AMET-PHYSICS HORIZONS is the official magazine of the Physics Department at AMET University, highlighting the department's growth and faculty achievements during the odd semester. This issue covers the academic contributions, research, and professional development of our faculty members, along with their involvement in various workshops and seminars. It also celebrates the department's progress, fostering a culture of innovation and excellence in the field of Physics. Through this magazine, we reflect on our collective achievements and the department's continued success.

Chancellor

Dr. J. Ramachandran

President

Dr. Rajesh Ramachandran

Vice-Chancellor

Prof. Dr. V. Rajendran

Provost

Col.Dr.G.Thiruvassagam

Editor-in-Chief

Dr. K. Anandan, Associate Professor and Head
Department of Physics, AMET Deemed to be University

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FOREWORD



It is with great pleasure that I present this edition of the Physics Department's Academic year 2024-25 Even Semester Magazine, a reflection of the continued growth and vibrant academic culture we foster at AMET University. As a department dedicated to preparing the next generation of leaders in technical and maritime fields, our primary goal remains to equip undergraduate students with not only a solid understanding of fundamental physics but also the practical skills and critical thinking required to excel in their respective careers. Our faculty members, who are passionate about both teaching and research, continue to make remarkable strides in a wide range of scientific domains. Through their active participation in international conferences, workshops, and FDPs, they bring the latest advancements and methodologies into our classrooms. Their involvement in research keeps our department at the forefront of innovation, and it is through these scholarly pursuits that we aim to inspire and cultivate a deeper interest in the sciences among our research students.

This semester, our department has continued to uphold its tradition of organizing events that contribute to both academic and professional development. From seminars on research methodology, intellectual property rights, skill development programs and guest lectures from industry experts, we remain committed to creating opportunities for our students to engage with emerging trends and technologies. Additionally, our involvement in submitting project proposals to government agencies ensures that our faculty members are exposed to real-world problem-solving and research applications. I am confident that the dedication, enthusiasm, and intellectual curiosity demonstrated by our faculty and students will continue to elevate the standards of our department. I am proud of the commitment to excellence demonstrated by everyone involved and look forward to the continued growth and achievements of our department. I hope that this magazine provides a comprehensive overview of the ongoing activities, research, and milestones of the Physics Department. I extend my best wishes to our students and faculty as we continue to move forward together, striving for academic excellence and innovation.

Dr. K. Anandan
Head of the Department

PREAMBLE

It gives us immense pleasure to present the magazine of Physics department, AMET University a publication that reflects the academic spirit, research excellence, and creative talents of our faculty members. This magazine highlights the department's activities, faculty's initiatives, contributions, and achievements that continue to strengthen our commitment to quality education and scientific advancement.

The Department of Physics, established in 2001–2002, plays a vital role in laying a strong foundation in science and technology. With a dedicated team of ten faculty members holding PhDs and post-doctoral experience from reputed national and international institutions, the department offers Postgraduate and Doctoral programmes designed to prepare students for careers in frontier areas of science. Our research is supported by leading national funding agencies such as DST, CSIR, and DAE, and our faculty have contributed over 300 publications in prestigious journals.

As part of **AMET University—India's first Maritime Deemed to be University** our department contributes to a unique academic environment that combines maritime focus with strong grounding in basic and applied sciences. AMET, recognized globally and accredited with **NAAC 'A' Grade**, is a hub of maritime education and innovation, serving over 3500 students across diverse faculties. The university's strong ties with global industries, international collaborations, and state-of-the-art facilities provide a vibrant platform for holistic student development. This magazine is a testament to our department's continuous efforts to promote scientific inquiry, interdisciplinary collaboration, and academic excellence. We sincerely thank all the contributors, coordinators, and editorial team members for their valuable inputs and efforts in bringing this publication to fruition.

VISION AND MISSION OF THE UNIVERSITY

VISION

To sustain identity as a World Class Leader in Maritime Education and empower learners with wholesome knowledge through progressive innovation in training, research and development which will render students a unique learning experience and a transformation impact on the Global Society.

MISSION

AMET will strive continuously to

- ❖ Impart value-based higher education and technical knowledge with uncompromising strides of an outstanding quality.
- ❖ Emerge as a Centre of Excellence inculcating skill development in recent technologies in accordance with industrial trends.
- ❖ Create World class research capabilities on par with the finest in the world and broaden student's horizons beyond classroom education.
- ❖ Nurture talent and entrepreneurship to enable all round personality development among students.
- ❖ Empower students across socio economic strata.
- ❖ Make a positive difference to society through technical education.

QUALITY POLICY

AMET is committed to provide the highest quality in education and be the most preferred institution for pursuing marine and marine related courses.

This will be achieved by consistent focus on:

- ❖ Providing a conducive, vibrant, progressive and enriching learning atmosphere.
- ❖ Teaching Excellence and Research Output.
- ❖ Global outlook and engaging with the world through learning, teaching and research.
- ❖ Providing competitive advantage in gaining employment for further academic opportunities.
- ❖ Maintaining excellent links with commerce and industry both national and international.
- ❖ Complying with all applicable requirements and continually improving the effectiveness of Quality Management System.

VISION AND MISSION OF THE DEPARTMENT

VISION

Our primary responsibility is to prepare undergraduate students for careers in their respective technical fields. In light of the fact that Physics is a bridge between technology and society, the Department of Physics is also committed to bringing an understanding of physics in particular and science in general to both the general student population and the campus community at large.

MISSION

The Department of Physics provides high quality Physics education; producing well prepared graduates. The department of Physics promotes research and creative activities of faculty members and students.

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LIST OF FACULTY MEMBERS



Dr. Anita R Warriar.
Associate Professor & Dean

Dr. K. Anandan
Associate Professor & Head



Dr. A. Mohamed Haroon Basha
Associate Professor

Dr. K. Rajesh
Associate Professor



Dr. K. Gayathri
Associate Professor and
Dy.Head

Dr. Anitha Rexalin Devaraj
Associate Professor



Dr. G. Jayalakshmi
Associate Professor

Dr. C. Amirthakumar
Assistant Professor



Dr. Swapna Mary Gali
Assistant Professor

Dr. T.U. Jeevitha
Assistant Professor



Mr. M. Kamalakkannan
Lab Instructor

Mrs.T.Saranya
Lab Instructor



Mrs. P. Gomathi
Department Secretary

COURSE OFFERED – EVEN SEMESTER

S. No.	Course	Program
1	Nautical Physics and Electronics – II	B.Sc., Nautical Science
2	Nautical Physics and Electronics Lab – II	B.Sc., Nautical Science
3	Engineering Physics	B.E., - Mechanical Engineering
4	Engineering Physics Lab	B.E., - Mechanical Engineering
5	Applied Physics-II	B.Sc.,- Aviation
6	Next Generation – Luminescent Materials	B.Sc., and B.E (VAC)
7	Solar Energy Generation	B.Sc., and B.E (VAC)
8	Plastic Technology and their Sustainability	B.Sc., and B.E (VAC)
9	Industrial Wastewater Treatment	B.Sc., and B.E (VAC)

AWARDS AND RECOGNITIONS



Dr. Anita R. Warriar, Professor and Dean (S&H), received the *People's Choice Award* in the *Women in Space Leadership Programme (WiSLP)*, organized by the Department of Science and Technology, held at the **Indian Association for the Cultivation of Science (IACS)**, Kolkata, during January 2025.



Reviewer Recognitions from ELSEVIER and Springer Nature



WILEY Best Poster Award

- i) Dr. K. Anandan, has been awarded the **Reviewer Recognition Award** by **Elsevier** for his valuable peer-review contributions to the journal *Colloids and Surfaces A: Physicochemical and Engineering Aspects*. This recognition, along with similar acknowledgements from **Springer Nature journals** (*Discover Applied Science*, *Ionics*, and *Catalysis Letters*), reflects his dedication to advancing scientific excellence. The department extends heartfelt congratulations on this academic honour.
- ii) Dr. K. Anandan, Associate Professor and Head of the Department, has been awarded the **Elite Silver Award** for successfully completing the **12-week NPTEL-AICTE online course** titled “*Physico-Chemical Processes for Wastewater Treatment*”.
- iii) Dr. K. Anandan, received the **WILEY Best Poster Award** with a shield and cash prize at the **8th International Conference on Nanoscience and Nanotechnology (ICONN-2025)** held at SRM Institute of Science and Technology, Chennai, from **24th to 26th March 2025**. This recognition reflects his innovative contributions in the field of nanoscience. The department proudly congratulates Dr. Anandan on this remarkable achievement.



Reviewer Recognitions from ELSEVIER and IJMSE & NPTEL Elite

- iv) Dr. K. Rajesh, has been awarded the **Reviewer Recognition Award** for his valuable peer-review contributions to the journals **IJMSE**, reflects his dedication to advancing scientific excellence. The department extends heartfelt congratulations on this academic honour.
- v) Dr. K. Rajesh, Associate Professor, has been awarded the **Elite Silver Award** for successfully completing the **4-week NPTEL-AICTE online course** titled “*Outcome based pedagogic principles for effective teaching*”.



Dr. K. Gayathri, Associate Professor and Deputy Head, was **invited as a guest speaker** at **Sai Ram Engineering College**, where she delivered a lecture on the topic **"Crystal Growth and Synthesis Techniques"** on **4th January 2025**. Her expert insights on advanced material synthesis were well-received and appreciated by the academic audience.



Dr. Anitha Rexalin Devaraj, Associate Professor, has received **Reviewer Recognition** from several reputed international journals, acknowledging her valuable contributions to the peer-review process. Her recognitions include:

- **BP International**
- **Asian Journal of Research in Biochemistry**
- **Journal of Scientific Research and Reports**
- **Asian Journal of Chemical Sciences**
- **Journal of Applied Physical Science International**

Her dedicated service as a reviewer reflects her academic excellence and commitment to upholding the quality of scientific publications.



Dr. Anitha Rexalin Devaraj, Associate Professor and Director – AMET Institutions' Innovation Council, served as an **Evaluator** for the **School Innovation Marathon (2024–25)**, organized by the **Ministry of Education's Innovation Cell**, in collaboration with the **Department of School Education & Literacy** and **Atal Innovation Mission**. Her contribution supported the promotion of innovation among school students nationwide.



Dr. Anitha Rexalin Devaraj, Associate Professor and Director, AMET-IIC has received **“PEOPLE’S CHOICE AWARD”** in **Four-day leadership training program under the Women in Space and Allied Science Leadership Program (WisLP)** sponsored by the Department of Science and Technology (DST), Government of India, New Delhi, in collaboration with the British Council as part of the UK-India Education and Research Initiative (UKIERI) (18.02.2025-21.02.2025) held at Indian Institute of Technology, Indore



Dr. C. Amirthakumar, Assistant Professor of Physics, received the **Best Paper Presentation Award** at the **International Conference on Energy Materials (ICOEM – 2025)**, held on 17th February 2025 at Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603110.

PATENTS

Office of the Controller General of Patents, Designs & Trade Marks
Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India
(http://ipindia.nic.in/index.htm)

INTELLECTUAL PROPERTY INDIA
INNOVATION & PROTECTION

(http://ipindia.nic.in/index.htm)

Application Details	
APPLICATION NUMBER	202441099840
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	17/12/2024
APPLICANT NAME	1. Dr. K. Anandan 2. Mr. Kanchanapala Varma Reddy 3. Mr. Talsala Venkatesh 4. Mrs. R. Gayatri 5. Dr. AZEEZA V 6. Dr. R. Premalatha 7. Dr. Subarna Kumar Swain 8. Dr. A. UCHAKUMAR 9. Mr. N. Mahalingam 10. HEMANT KUMAR URADHUR
TITLE OF INVENTION	Innovations in high-density lithium-ion batteries and efficient wireless power transfer systems for EVs
FIELD OF INVENTION	ELECTRICAL
E-MAIL (As Per Record)	patent@icard@gmail.com
ADDITIONAL E-MAIL (As Per Record)	padhu003@gmail.com
E-MAIL (UPDATED Online)	
PRIORITY DATE	—
REQUEST FOR EXAMINATION DATE	—
PUBLICATION DATE (BUS 11A)	03/01/2025

Dr. K. Anandan, Associate Professor and Head of the Department, has published a patent titled **"Innovation in High-Density Lithium-Ion Batteries and Efficient Wireless Power Transfer Systems for Electric Vehicles"** (Application No. **202441099840A**, Publication Date: **03.01.2025**).

Office of the Controller General of Patents, Designs & Trade Marks
Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India
(http://ipindia.nic.in/index.htm)

INTELLECTUAL PROPERTY INDIA
INNOVATION & PROTECTION

(http://ipindia.nic.in/index.htm)

Application Details	
APPLICATION NUMBER	202521045743
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	13/05/2025
APPLICANT NAME	1. Dr. Rishi Pandey 2. Ms. D. Srihar 3. Mr. Mohanlal Akkari 4. Dr. P. Devendran 5. Dr. Ananda A 6. Dr. SANGEET C 7. Dr. K. Rajesh 8. DR. S. THEJASWARI KAVESANUMAR 9. Mr. Anuj Kumar Dandia 10. Mrs. Muskan V
TITLE OF INVENTION	Self-Healing Concrete Using Nanomaterials to Extend Infrastructure Longevity
FIELD OF INVENTION	POLYMER TECHNOLOGY
E-MAIL (As Per Record)	patent@icard@gmail.com
ADDITIONAL E-MAIL (As Per Record)	patent@icard1@gmail.com
E-MAIL (UPDATED Online)	
PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	—
PUBLICATION DATE (BUS 11A)	30/05/2025

Dr. K. Rajesh, Associate Professor, has published a patent titled **'Self-Healing Concrete Using Nanomaterials to Extend Infrastructure Longevity'**. (Application No. **202521045743** Publication Date: **30.05.2025**).

Office of the Controller General of Patents, Designs & Trade Marks
Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India
(http://ipindia.nic.in/index.htm)

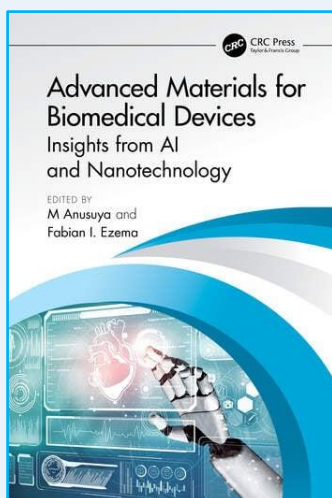
INTELLECTUAL PROPERTY INDIA
INNOVATION & PROTECTION

(http://ipindia.nic.in/index.htm)

Application Details	
APPLICATION NUMBER	202441099839
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	17/12/2024
APPLICANT NAME	1. Dr. Anitha Rexalin Devaraj 2. Dr. Kolla Naga Sreenivasa Rao 3. Dr. Ananda A 4. Prof. L. Venkata R 5. Dr. Gresh K M 6. Anuram Mantharappayath 7. Saja Sumina anaz 8. Dr. Dhruva Kumar 9. Mr. T. Koushikdasan 10. Dr. Vikram Koudamudi Venu
TITLE OF INVENTION	Nanotechnology-based methods for developing efficient water purification systems
FIELD OF INVENTION	CHEMICAL
E-MAIL (As Per Record)	patent@icard@gmail.com
ADDITIONAL E-MAIL (As Per Record)	padhu003@gmail.com
E-MAIL (UPDATED Online)	
PRIORITY DATE	—
REQUEST FOR EXAMINATION DATE	—
PUBLICATION DATE (BUS 11A)	03/01/2025

Dr. Anitha Rexalin Devaraj, Associate Professor and Director, AMET-IIC, has published a patent titled **"Nanotechnology-based methods for developing efficient water purification systems."** (Application No. **202441099839**, Publication Date: **03.01.2025**).

BOOK/BOOK CHAPTERS PUBLISHED



Publication in International Book

1. **Dr. K. Anandan**, Associate Professor and Head of the Department, has contributed **two book chapters** to the internationally published book titled "*Advanced Materials for Biomedical Devices*," by **CRC Press, Taylor & Francis Group** (ISBN: 9781041096931).

- ❖ **Chapter 16:** *Renewable Energy Storage Systems Using Advanced Materials for Electric Vehicle and Biomedical Applications with AI-Driven Predictive Energy Management*
- ❖ **Chapter 28:** *Challenges in Scaling Nanotechnology-Based Materials for Large-Scale Industrial and Biomedical Applications Using AI-Powered Quality Control Systems*

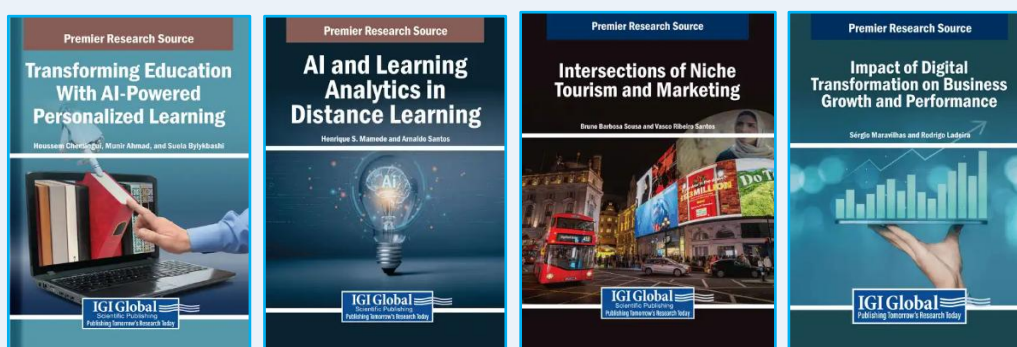
These contributions reflect his expertise at the intersection of **nanotechnology, AI, and energy systems**. The department congratulates Dr. Anandan for his continued academic excellence and impactful research dissemination.

2. **Dr. K. Rajesh**, Associate Professor, has contributed **Two book chapters** to the internationally published book titled "*Advanced Materials for Biomedical Devices*," in by **CRC Press, Taylor & Francis Group** (ISBN: 9781041096931).

- ❖ **Chapter 32:** *Surface Engineering and Nanostructuring for Enhanced Mechanical, Thermal, and Biocompatibility Properties in Industrial and Biomedical Components Using Computational Modeling*
- ❖ **Chapter 40:** *Biocompatible Materials and Smart Polymers for Targeted Biomedical Applications Using AI for Material Behavior Prediction and Nanotechnology Enhancements*

3. **Dr. K. Gayathri**, Associate Professor and Deputy Head of the Department, has contributed **two book chapters** to the internationally published book titled "*Advanced Materials for Biomedical Devices*," by **CRC Press, Taylor & Francis Group** (ISBN: 9781041096931).

- ❖ **Chapter 53:** *Nanotechnology-Enhanced Coatings for Corrosion Resistances in Marine, Industrial, and Biomedical Engineering Leveraging AI for Predictive Maintenance.*
- ❖ **Chapter 55:** *Hybrid Nanocomposites for High-Performance Applications in Aerospace, Mechanical, and Biomedical Engineering Enhanced by Computational Modelling and AI*



Dr. K. Gayathri, Associate Professor and Deputy Head, Department of Physics, for her remarkable scholarly contributions through book chapter publications in reputed international volumes. Her achievements bring great recognition to our Department and University.

1. **Book:** *Transforming Education with AI-Powered Personalized Learning*
Chapter: *Fostering Personalized Learning and Achieving Equity in Education: The Role of AI-Powered Curriculum Development*
Publisher: IGI Global Scientific Publishing
DOI: 10.4018/979-8-3693-8382-7.ch008
2. **Book:** *AI and Learning Analytics in Distance Learning*
Chapter: *Role of Digital Marketing in Shaping Travel Decisions: Consumer Behavior in Tourism*
Publisher: IGI Global Scientific Publishing
DOI: 10.4018/979-8-3693-8417-6.ch007
3. **Book:** *AI and Learning Analytics in Distance Learning*
Chapter: *Creating Engaging and Personalized Learning Experiences in Distance Education: AI and Learning Analytics*
Publisher: IGI Global Scientific Publishing
DOI: 10.4018/979-8-3693-7195-4.ch005
4. **Book:** *Impact of Digital Transformation on Business Growth and Performance*
Chapter: *Leveraging Machine Learning for Enhanced Patient Engagement and Outcomes: Revolutionizing Healthcare Marketing*
Publisher: IGI Global Scientific Publishing
DOI: 10.4018/979-8-3693-9783-1.ch012

4. **Dr. Anitha Rexalin Devaraj**, Associate Professor and Director, AMET-IIC, has contributed **two book chapters** to the internationally published book titled "*Advanced Materials for Biomedical Devices*," by **CRC Press, Taylor & Francis Group** (ISBN: 9781041096931).

❖ **Chapter 22:** *Biodegradable and Eco-Friendly Materials for Sustainable Construction and Biomedical Packaging Applications Developed Using AI-Driven Life Cycle Assessments*

❖ **Chapter 34:** *Advanced Fluid Dynamics in Microfluidic Systems for Environmental, Chemical, and Biomedical Engineering Applications Simulated Using Computational Tools*

Her contributions underscore a strong expertise spanning **biomedical applications, artificial intelligence, and energy systems**. The department extends its heartfelt congratulations to Dr. Anitha Rexalin Devaraj for her ongoing academic excellence and meaningful research contributions.



5. **Dr. A. Mohamed Haroon Basha, Associate Professor**, has contributed book chapters to the internationally published book titled "*Challenges of Sustainable Development*", published by **RFI Publications** (ISBN: 978-81-19135-27-1). **Chapter 10:** *Environmental Challenges to Sustainable Development*

Dr. A. Mohamed Haroon Basha, Associate Professor, has contributed another book chapter to the internationally published book titled "*Contemporary Developments in Science and Technology*", published by **Bhumi Publishing, India** (ISBN: 978-93-48620-02-6). **Chapter 04:** *Innovations in Perovskite Solar Cells Driving High-Performance Energy Storage*

RESEARCH ARTICLE PUBLICATIONS

Our department faculty members are actively engaged in cutting-edge research across a wide range of scientific fields. As a result of their dedication and scholarly efforts, they have published numerous research articles in reputed international journals. These publications not only reflect the high academic standards of our department but also contribute significantly to the advancement of knowledge in their respective areas.



Dr. K. Anandan, Associate Professor and Head of the Department, has published a research article in the **Chemosensors** journal (Impact Factor: 3.7), a reputed international journal in the field of sensor technology and materials science.



Dr. A. Mohamed Haroon Basha, Associate Professor, has published two research articles in the **International Journal of Environmental Science** and **Journal of Neonatal Surgery**.

Nonenzymatic Selective Electrochemical Determination of Dopamine Using Fe³⁺-Doped Hydroxyapatite

Korinjithan Pannervelam, Anila R. Warrier, Roselin Rajitha Mathiasan, Vijayaraj Radha, and Rajesh Krishnan

1 Introduction

Dopamine (DA) is a neurotransmitter belonging to the catecholamine family, serving as both a biological hormone and a neurotransmitter. The sensory and central nervous systems of humans rely on it to play a vital role in sustaining the metabolic activity of the entire human body [Kim et al., 2018; Gong et al., 2020; Arumugam et al., 2020]. DA levels have an essential role in the body which regulates several physiological processes including attention, retention, learning, mobility behavior, mental awareness, personality, etc. Elevated quantities of DA have been seen to potentially correlate with many neuropsychiatric disorders mainly Alzheimer's,

The original version of the chapter has been revised. A correction to this chapter can be found at https://doi.org/10.1007/978-3-311-69970-2_2

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Spectral Analysis and Biological Evaluation of 3,5-Dimethyl-1-[2-(1-methyltetrahydro-5-yl)-sulfonyl]acetyl-2,6-diphenylpiperidin-4-one: Crystal Structure, Hirshfeld Surface, Antimicrobial Activity and Dipeptidyl Peptidase IV Inhibition

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A single crystal of 3,5-dimethyl-1-[2-(1-methyltetrahydro-5-yl)-sulfonyl]acetyl-2,6-diphenylpiperidin-4-one (DMTHSPPO) has been synthesized and extensively characterized using analytical techniques, which include: ¹H NMR, infrared, mass spectroscopy and single-crystal X-ray diffraction (SCXRD). DMTHSPPO is a colorless, crystalline solid with a molecular weight of 330.41 g/mol, C₂₀H₂₄N₂O₄ (C₂₀H₂₄N₂O₄) and a melting point of 149–150 °C. The molecular structure was identified through a combination of methods and refined with SHELXL-97, which produced a final R-value of 0.0486. The Hirshfeld surface analysis has been employed to determine the nature of intermolecular interactions between molecules, revealing predominant contributions from C–H and N–H intermolecular contacts. The derivatives were subjected to *in vitro* antimicrobial assays against bacterial and fungal strains, with DMTHSPPO exhibiting significant activity against *E. coli*. The release mechanism of action for piperidine derivatives prompted a molecular docking study, showing high affinity for the active site region, binding modes and interactions of active site residues with dipeptidyl peptidase IV (DPP-IV). This investigation contributes to a detailed understanding of the compounds' mechanism of action and holds promise for the design of novel, potent drug molecules.

Keywords: Density functional theory, Hirshfeld surface evaluation, Molecular docking, Dipeptidyl, Single crystals.

INTRODUCTION

Nitrogen-containing heterocyclic molecules, particularly piperidine-ones compounds, have gained importance due to their miscellaneous biological characteristics, which include antiviral, anticancer, antitumor [1], analgesic, fungicidal [2], local anesthetic, antitumor [3,4], anti-inflammatory, antimicrobial, bactericidal, herbicidal, antidiabetic, CNS stimulant and antidepressant activities [5–7]. Compounds with an amide bond linkage have exhibited an extensive variety of biological activities, which include antitumor activities [8]. The synthesis of naphthalene [9], antitumor and general anesthetics [10], antidiabetic medicines, narcotic analgesics, transmitters and hypertensive drugs all include a piperidine ring [11]. Various piperidine derivatives refer to the skeletons for a variety of

alkaloids [12]. The piperidine heterocycle primarily exists in chair conformation [13,14], but depending on the replacement, its arrangement and conformation may change [15,16].

Piperidines also perform an essential part in medicine since they are necessary chiral intermediate substances in the synthesis of a wide range of synthetic, natural and semisynthetic pharmacological agents with effective cancer prevention and anti-IV characteristics [17]. Transplantation piperidine compounds should be evaluated for effectiveness in immunosuppressing animals [18]. Nitrogen heterocycles with dipeptidyl substrate activity have antiproteolytic activity [19–22]. The synthesis of α -piperidones is currently being studied because of its potential therapeutic applications [23–24]. Similar to other alkaloids, like α -piperidones were found in lower blood cholesterol levels [25,26]. Arlowski et al. [27–31] studied the antibacterial, anal-

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Dr. K. Rajesh, Associate Professor, has published two research articles in reputed international journals, including the **Springer Proceedings in Physics**, a well-regarded journal in the field of Physics published by Springer, and the **Asian Journal of Chemistry**, a respected journal in the field of Structural Chemistry.

Development and Characterization of Gadolinium-Doped Hydroxyapatite to Enhance Biocompatibility in Biomedical Applications

Swadhi Radhakrishnan¹, Gayathri Krishnan², Anutha Rexalin Devaraj³, Rajesh Krishnan, Anandan Kasinatha⁴

¹Department of Physics, Academy of Maritime Education and Training, Chennai, India

Abstract

The synthesis and characterization of gadolinium-doped brucinium hydroxyapatite (Gd-BHAP) have garnered significant attention for their potential applications in bone regeneration and dentistry. Hydroxyapatite (HAP) closely resembles the mineral phase of bone, exhibiting biocompatibility, biodegradability, and bioactivity. Hence, this study aims to synthesize and characterize Gd-BHAP to evaluate its potential for enhancing biocompatibility and effectiveness in bone regeneration applications. Gd-BHAP was synthesized via a hydrothermal method using calcium nitrate tetrahydrate and diammonium hydrogen phosphate as precursors, maintaining a calcium-to-phosphate molar ratio of 1.6. Dual dopants, brucine (1%) and gadolinium (~0.5%, 1%, and 2% concentrations), were incorporated, with the pH adjusted to 9. Characterization was performed using XRD, SEM, DLS, FTIR, Fluorescence and UV-Vis spectroscopy. The cytotoxicity of Gd-BHAP was evaluated on Terc cell viability using the MTT assay across varying concentrations (25–200 µg/mL). Results indicated that the characterization techniques confirmed the successful synthesis of Gd-BHAP, demonstrating rod-shaped morphology and sizes ranging from 70 to 121 nm. FTIR analysis revealed typical absorption bands of hydroxyapatite, while XRD patterns matched known standards for HA, indicating a hexagonal phase. Cytotoxicity results showed a concentration-dependent decrease in Terc cell viability, with 84% viability at 25 µg/mL, decreasing to 73% at 200 µg/mL, indicating acceptable biocompatibility for potential biomedical applications. In conclusion, the synthesized gadolinium-doped brucinium hydroxyapatite exhibits promising characteristics for biomedical applications, particularly in bone regeneration. While higher concentrations may reduce cell viability, the material demonstrates significant biocompatibility, positioning it as a suitable candidate for tissue engineering.

Keywords: Biocompatibility, Biocompatibility, Biomedical, Gadolinium-Doped Brucinium Hydroxyapatite, Hydroxyapatite

Introduction

The development of nanosized carriers for drug delivery to tumor sites is one of the most active fields in nanomedicine [1]. Constructing drug carriers with excellent biocompatibility is crucial for the clinical translation of these nanosystems [2, 3]. In recent years, hydroxyapatite (HAP), a calcium phosphate bioceramic, has gained attention as a building block for nanosized drug carriers due to its bioactivity, non-toxic nature, and non-inflammatory properties [4, 5, 6]. HAP is well-known for its suitability in bone tissue regeneration [7], owing to its biocompatibility [8, 9], good mechanical properties [10], osteoconductive [11, 12], and ability to form chemical bonds with bone tissue [13, 14]. This ceramic has been utilized for treating bone defects [10, 15] however, the mechanical strength of most hydroxyapatite scaffolds

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Articles

Magnesium-Doped Brucinium Hydroxyapatite: A Versatile Material for Biomedical Applications

Swadhi Radhakrishnan¹, Gayathri Krishnan², Anutha Rexalin Devaraj³, Rajesh Krishnan⁴, Anandan Kasinatha⁵

¹Research Scholar, Department of Physics, Academy of Maritime Education and Training, Chennai, India.

^{2,3,4,5} Associate Professor, Department of Physics, Academy of Maritime Education and Training, Chennai, India.

Abstract:

Hydroxyapatite being a biocompatible material, which closely resembles the mineral phase of bone, it has been used for various biomedical application such as implantation, augmentation and drug delivery etc. Magnesium being a potential trace element towards the enhancement of physicochemical properties of HAP and brucine being an effective traditional medicine extracted from plant alkaloid, together promoter to the effective biocompatible for various applications. A dual dopant: Magnesium and Brucine incorporated with hydroxyapatite nanoparticles (Mg-BHAP) were prepared by a hydrothermal method using calcium nitrate and diammonium phosphate as precursors. The characterization indicates the successful synthesis of Mg-BHAP, demonstrating clamped-shaped morphology and sizes ranging from 80 to 130 nm. FTIR analysis revealed typical absorption bands of hydroxyapatite, while XRD patterns matched known standards for HA, indicating a hexagonal phase. The UV-Vis-NIR optical absorptions study highlights the electronic and structural modification due to the dopants: confirm the potential application in the field of optical filters and bio-photonics.

Keywords: Hydroxyapatite, dual dopant, trace element, plant alkaloid, FTIR, XRD, SEM, UV, biomedical.

1. Introduction:

Hydroxyapatite (HAP) is a calcium phosphate material with the chemical formula of Ca₁₀(PO₄)₆(OH)₂ which has the stoichiometry Ca/Po ratio as 1.67 makes a vital compound in the area of biomedical research for the biocompatible, osteoconductive, chemical stability seamlessly integrate with the tissues, supporting an extensive range of application in regenerative medicine and tissue engineering. HAP yields more attention for the resemblance of mineral compound of human bone & teeth. It employs in the bone grafting, scaffold, implantation coating to encourage osseointegration optimizing the performance and longevity of prosthetics [1]. Also exploited in the field of dental for its dental filler, cement and remineralization therapies owing the potential to adhering with

Guest feature 2025 5(4):288-290

Dr. K. Gayathri, Associate Professor and Deputy Head, has actively contributed to the advancement of scientific knowledge through her research publications and leadership participation. She has published **two research articles** in reputed international journals: *Texila International Journal of Public Health* and *Cuestiones de Fisioterapia*, reflecting her commitment to impactful and interdisciplinary research.

Dr. G. Jayalakshmi, Associate Professor of Physics, has published two significant research articles in reputed international journals (ELSEVIER):

1. “Top-down approach for the preparation of Au/ZnO nanostructures by glancing-angle ion irradiation: Morphological, structural and optical studies,” **Journal of Luminescence**, Volume 277, January 2025, 120974.
2. “Tunable visible-light photodetection in un-doped ZnO thin films by Xe⁺ ion irradiation,” **Sensors and Actuators A: Physical**, Volume 391, 1 September 2025, 116617.

These publications reflect her consistent research excellence and valuable contributions to the field of nanomaterials and optoelectronic applications. The Department and Management wholeheartedly appreciate her achievements.



Dr. C. Amirthakumar, Assistant Professor of Physics, has published a research article in the journal **Cuest. Fisioter** titled “Eco-Friendly Synthesis of Water-Soluble Imidazole Derivatives: Exploring Multiple Pathways and Their Catalytic Properties.” This achievement reflects his dedication to advancing sustainable chemical research and is highly commendable.



Dr. Swapna Mary Gali, Assistant Professor, has published research articles in reputed international journals, including the **Asian Journal of Chemistry** (Impact Factor: 0.73) and the **Journal of The Electrochemical Society** (Impact Factor: 3.3).



Dr. T. U. Jeevitha, Assistant Professor, has published a research article titled “A Study on the Trajectories of Test Particles Surrounding Naked Singularities in the Bertrand Spacetime-II” in the **AIP Conference Proceedings**, a reputed international journal published by the American Institute of Physics (Scopus indexed). The paper contributes to advancements in general relativity and geodesic motion around singularities.

E-CONTENT DEVELOPMENT

This series of video lectures provides in-depth knowledge on essential topics in Physics. Each lecture is delivered by a faculty member from the Department of Physics, covering fundamental concepts and real-world applications to enhance learning.

Name and Designation	Topics
Dr. Anita R. Warriar Professor and Dean (S&H)	❖ Journey from Nano to Meta World – An overview of the evolution from nanomaterials to emerging metamaterials. ❖ Nanotechnology – Study of materials and devices at the nanoscale for various scientific applications. ❖ Semiconductor Nanoparticles – Nanostructures with tunable electronic and optical properties used in sensors and solar cells. ❖ Ancient Nano World – Historical use of nanomaterials in ancient artifacts and practices. ❖ Indium Sulphide Nanoparticles – A promising semiconductor material with applications in photocatalysis and optoelectronics. ❖ Discoveries and Inventions with Nanomaterials – Breakthrough innovations enabled by the unique properties of nanomaterials.
Dr. K. Anandan Associate Professor and Head	❖ Magnetic Effect of Electric Current – Electric current produces a magnetic field, forming the basis of electromagnetism. ❖ Transmission of Heat and Heating Effect of Electric Current – Understanding heat flow and heat generation due to electric current. ❖ Li-Ion Battery – A rechargeable energy storage device widely used in electronics and electric vehicles.
Dr. K. Rajesh Associate Professor	❖ Transformer – A device used to change voltage levels in alternating current circuits. ❖ Losses in the Transformer – Various types of energy losses occurring in transformers during operation. ❖ Heat – A form of energy transfer between bodies due to temperature difference
Dr. K. Gayathri Associate Professor and Deputy Head	❖ Communication Techniques – Methods used to transmit and receive information effectively between a sender and receiver. ❖ Satellite Communication – Transmission of signals using artificial satellites for long-distance communication. ❖ Satellite Orbits – The specific paths followed by satellites around the Earth or other celestial bodies. ❖ Orbital Perturbation – Small changes in a satellite's orbit caused by gravitational or environmental forces. ❖ Why we need Modulation – Modulation allows signals to travel long distances efficiently without loss of quality. ❖ Multiplexing and its Types – A technique that combines multiple signals into one channel, with types like TDM, FDM, and CDM.
Dr. Anitha Rexalin Devaraj Associate Professor	❖ Thickness of Butter Paper using Air Wedge – A technique using interference to measure very thin film thickness. ❖ Rest and Motion – Fundamental concepts in mechanics describing the state of objects in space and time.

Dr. G. Jayalakshmi Associate Professor	❖ Introduction to Fiber Optics – The study of transmitting light through flexible glass or plastic fibers for communication. ❖ Nd:YAG Laser – Construction, Working and Applications – A solid-state laser used in medicine, industry, and research.
Dr. C. Amirthakumar Assistant Professor	❖ SONAR and its Applications – Use of sound waves to detect underwater objects and measure distances. ❖ Ultrasonic Waves and Ultrasonic Generation – High-frequency sound waves used in imaging, cleaning, and medical diagnostics.
Dr. Swapna Mary Gali Assistant Professor	❖ Importance of Physics in Daily Life – Physics explains natural phenomena and supports everyday technologies. ❖ Importance of Physics in Technology – Physics principles are foundational to technological advancements and innovations.
Dr. T. U. Jeevitha Assistant Professor	❖ De Broglie Wavelength – Concept showing that particles like electrons have wave-like properties. ❖ Nd:YAG Laser – Introduction to Solid-State Laser – A widely used solid-state laser with high efficiency and precision. ❖ CO₂ Laser – Introduction to Gas Laser – A powerful gas laser used in industrial cutting, welding, and medical surgery.



Dr. Anitha Rexalin Devaraj successfully completed transcription editing tasks using **Machine Learning Tools – Chitrlekha and TransLingua** for the **NPTEL-SWAYAM** courses **Computational Fluid Dynamics** and **Geosynthetics and Reinforced Soil Structures**, respectively, offered by **IIT Madras**, contributing a total of **9 hours** of editing.

FACULTY DEVELOPMENT PROGRAM

Commitment to Continuous Learning and Excellence

Our department faculty members actively participate in Faculty Development Programs (FDPs) to continuously enhance their teaching skills, stay updated with the latest advancements in pedagogy, and promote excellence in the learning process. These programs enable our faculty to integrate innovative teaching methodologies and modern educational tools, contributing to the overall academic growth of both educators and students.



Dr. K. Anandan, Associate Professor and Head of the Department, actively participated in the **7-Day International Online Faculty Development Programme** on "*Recent Developments and Applications in Advanced Materials*" held from **3rd to 9th March 2025**.

He also successfully completed a **12-week NPTEL-AICTE course** titled "*Physico-Chemical Processes for Wastewater Treatment*", securing **80% marks** and receiving the prestigious **Elite Silver Award**. These accomplishments reflect his dedication to academic advancement and professional growth. The department congratulates Dr. Anandan for his continued pursuit of excellence.



Dr. K. Rajesh, Associate Professor, actively participated in the **7-Day International Online Faculty Development Programme** on "*Recent Developments and Applications in Advanced Materials*" held from **3rd to 9th March 2025**.

He also successfully completed a **4-week NPTEL-AICTE course** titled "*Outcome based pedagogic principles for effective teaching*" securing **78% marks** and receiving the prestigious **Elite Silver Award**. These accomplishments reflect his dedication to academic advancement and professional growth. The department congratulates Dr. K. Rajesh for his continued pursuit of excellence.



Dr. K. Gayathri, Associate Professor and Deputy Head, actively participated in the **Intellectual Property UTSAV** event, held virtually from **21st to 26th April 2025**, organized by the **All India Council for Technical Education (AICTE)** in collaboration with the **Ministry of Education's Innovation Cell (MIC)**. The event focused on fostering awareness and capacity-building in intellectual property rights.

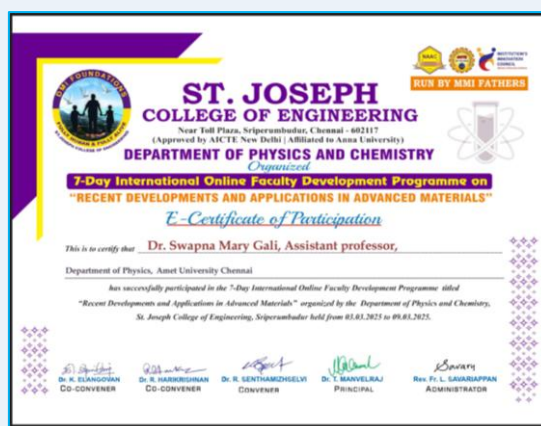
She also successfully completed a **7-Day International Online Faculty Development Programme** on "**Recent Developments and Applications in Advanced Materials**", conducted from **3rd to 9th March 2025**, enhancing her expertise in cutting-edge material science research and applications.



Dr. Anitha Rexalin Devaraj, Associate Professor, actively participated in the **Intellectual Property UTSAV** event, held virtually from **21st to 26th April 2025**, organized by the **All India Council for Technical Education (AICTE)** in collaboration with the **Ministry of Education's Innovation Cell (MIC)**. The event focused on fostering awareness and capacity-building in intellectual property rights.

Dr. Anitha Rexalin Devaraj, Associate Professor and Director, AMET-IIC, actively participated in the **5-Day Faculty Development Programme** on "**Innovation and Entrepreneurship**" held from **24th to 28th March 2025** at VIT, Chennai campus organised by All India Council for Technical Education (AICTE) & Ministry of Education's Innovation Cell (MIC).

She also participated in the **7-Day International Online Faculty Development Programme** on "**Recent Developments and Applications in Advanced Materials**" held from **03rd to 09th March 2025**. These accomplishments reflect her dedication to academic advancement and professional growth. The department congratulates Dr. Anitha Rexalin Devaraj for her continued pursuit of excellence.



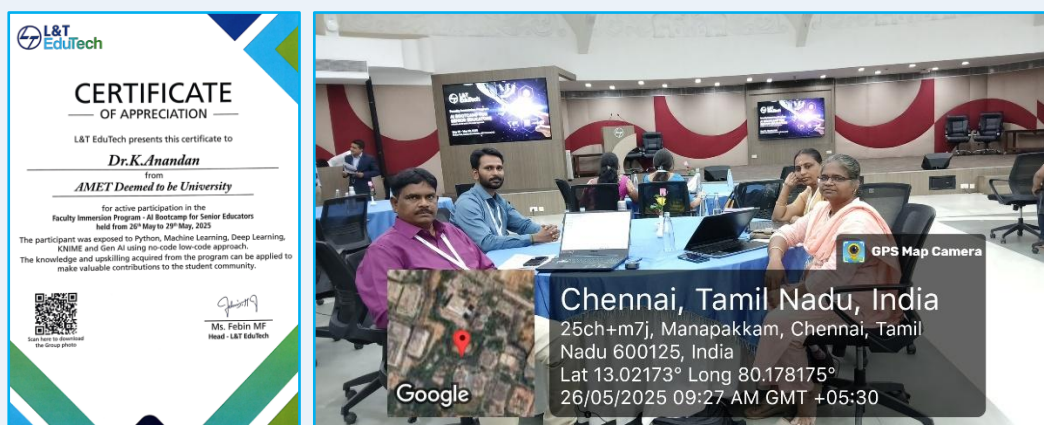
Dr. Swapna Mary Gali, Assistant Professor, actively participated in the **7-Day International Online Faculty Development Programme** on "*Recent Developments and Applications in Advanced Materials*" held from **3rd to 9th March 2025**.



Dr. T. U. Jeevitha, Assistant Professor, actively participated in two significant Faculty Development Programmes that reflect her commitment to professional growth and academic excellence.

She attended a **7-Day International Online FDP** on "*Recent Developments and Applications in Advanced Materials*" held from **3rd to 9th March 2025**, and also successfully completed a Faculty Development Programme conducted in **February 2025**, further enriching her knowledge in emerging areas of physics. The department congratulates **Dr. Jeevitha** for her active engagement in continuous learning and scholarly development.

FACULTY PROFESSIONAL DEVELOPMENT PROGRAM



Faculty Professional Development Program Participation

Dr. K. Anandan, Associate Professor and Head of the Department, attended the **Faculty Immersion Program – AI Bootcamp for Senior Educators** held from **26th to 29th May 2025** at **L&T EduTech, Chennai**. The program focused on enhancing AI competencies and digital teaching strategies for academic leaders.



Six faculty members of our Department successfully attended the **Training of Trainers (ToT) Program** on **"Innovative Pedagogical Approaches & Leadership Skills in the Industry 4.0 Era"**, organized by **NITTTR Chennai** from **2 to 6 June 2025** at **National institute of Technical Teachers Training and Research, Chennai**. The program aimed to enhance teaching innovation and academic leadership in the context of Industry 4.0.

Participated Faculties List

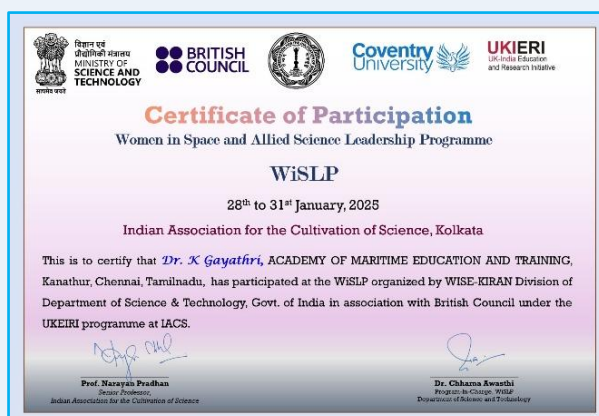
1. Dr. Anita R Warriar – Professor and Dean (S&H)
2. Dr. K. Rajesh – Associate Professor
3. Dr. K. Gayathri – Associate Professor
4. Dr. Anitha Rexalin Devaraj – Associate Professor
5. Dr. C. Amirthakumar – Assistant Professor
6. Dr. T.U. Jeevitha – Assistant Professor



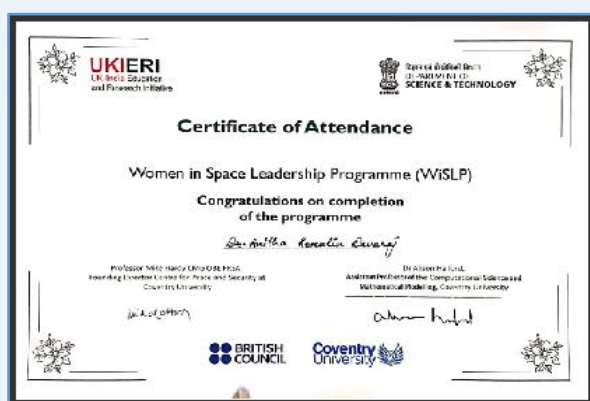
Four faculty members of our Department successfully attended the **Training of Trainers (ToT) Program** on “*Innovative Pedagogical Approaches & Leadership Skills in the Industry 4.0 Era*”, organized by **NITTTR Chennai** from 16-20th **June 2025** at **Sathyabama Institute of Science and Technology, Chennai**. The program aimed to enhance teaching innovation and academic leadership in the context of Industry 4.0.

Participated Faculties List

1. Dr. K. Anandan – Associate Professor and Head
2. Dr. A. Mohamed Haroon Basha – Associate Professor
3. Dr. G. Jayalakshmi – Associate Professor
4. Dr. Swapna Mary Gali – Assistant Professor



Dr. K. Gayathri, Associate Professor and Deputy Head, participated in the **Women in Space and Allied Science Leadership Programme** held from **28th to 31st January 2025**. The programme was organized by the **Department of Science and Technology (DST), Government of India**, in association with the **British Council**, under the **UK-India Education and Research Initiative (UKIERI)**, at the **Indian Association for the Cultivation of Science, Kolkata**.



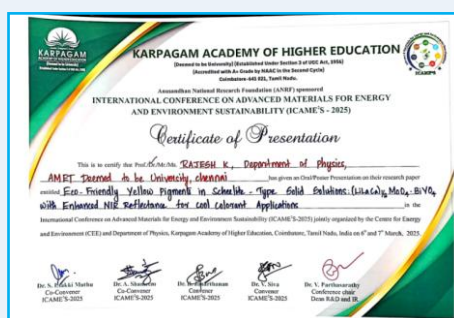
Dr. Anitha Rexalin Devaraj, Associate Professor and Director, AMET-IIC, received “**PEOPLE’S CHOICE AWARD**” in **Four-day leadership training program** under the **Women in Space and Allied Science Leadership Program (WisLP)** sponsored by the **Department of Science and Technology (DST)**, Government of India, New Delhi, in collaboration with the **British Council** as part of the **UK-India Education and Research Initiative (UKIERI)** (18.02.2025-21.02.2025) held at Indian Institute of Technology, Indore

NATIONAL AND INTERNATIONAL CONFERENCES ATTENDED

Our department faculty members actively participate in various international and national conferences to share their research findings, engage with the global academic community, and stay informed about the latest developments in their fields. By presenting their research outcomes at these prestigious platforms, our faculty not only contribute to the broader scientific dialogue but also gain valuable insights that further enrich their teaching and research practices.



Dr. K. Anandan, Associate Professor and Head of the Department, has actively participated and presented research papers in various international conferences, showcasing his innovative work in the fields of **nanomaterials**, **energy systems**, and **environmental applications**. His contributions reflect a strong commitment to global academic engagement and research dissemination.



Dr. K. Rajesh, Associate Professor, has actively participated and presented research papers in various international conferences, showcasing his innovative work in the fields of **nanomaterials**, **pigments**, and **environmental applications**. His contributions reflect a strong commitment to global academic engagement and research dissemination.



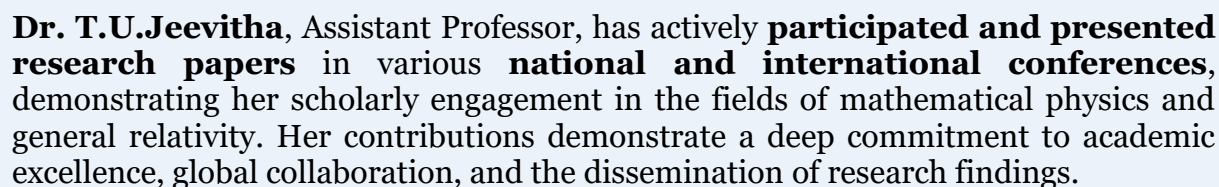
Dr. K. Gayathri, Associate Professor and Deputy Head, **presented a research paper** at the **4th International Conference on Multifunctional Materials and Radiation Measurements (ICMMRM-2025)**, held during **21st–22nd March 2025**. The conference was organized by **SSN College of Engineering, Chennai**, and brought together experts and researchers from across the globe to discuss advancements in multifunctional materials and radiation-related studies.



Dr. Anitha Rexalin Devaraj, Associate Professor and Director, AMET-IIC, has actively **participated and presented research papers** in various **international conferences**, showcasing her innovative work in the fields of **Density Functional Theory, and Spectral Analysis**. Her contributions reflect a strong commitment to global academic engagement and research dissemination.



Dr. C. Amirthakumar, Assistant Professor of Physics, has actively participated in several international conferences, where he presented his research papers showcasing innovative work. His contributions highlight a strong commitment to global academic engagement and the dissemination of research findings.



WORKSHOP ATTENDED



Dr. K. Anandan, Associate Professor and Head, attended a **Two-Day Workshop** on “*Structural Analysis using PXRD, Rietveld Refinement, HRTEM & SAED*” held on **24–25 February 2025** at **Sathyabama Institute of Science and Technology, Chennai**.

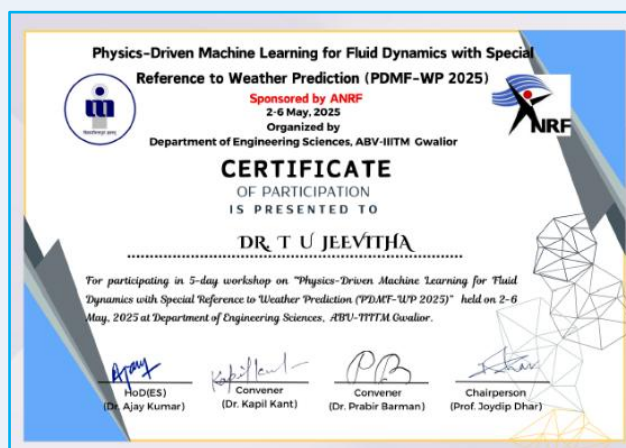
Dr. K. Rajesh, Associate Professor, attended a **Two-Day Workshop** on “*Advanced Transition metal Dichalcogenides for energy storage applications*” held on **11–12 March 2025** at **SSN Institution, Chennai**.



Dr. Anitha Rexalin Devaraj, Associate Professor and Director, AMET-IIC, attended a **Three-Days Science Academies' Lecture Workshop** on “*Density Functional Theory in Condensed systems : from clusters to solids*” held from **11–13 June 2025** at **St.Dominic's College, Kanjirapally, Kerala**.

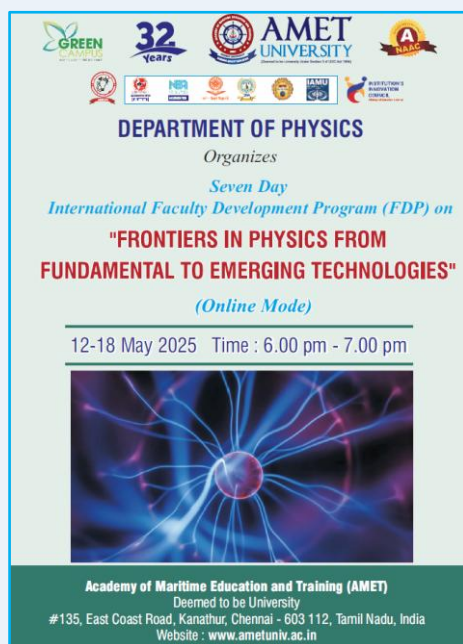


Dr. Swapna Mary Gali, Assistant Profess, attended a **Two-Day Workshop** on “**XRD and FESEM Analysis**” held on **27–28 May 2025** at **SSN college of Engineering, Chennai**.



Dr. T. U. Jeevitha, Assistant Professor, Department of Physics, attended a Five-Day Workshop on “**Physics-Driven Machine Learning for Fluid Dynamics with Special Reference to Weather Prediction (PDMF–WP 2025)**,” held from 2nd to 6th May 2025. The workshop was organized by the Department of Engineering Sciences, ABV–IIITM Gwalior, and sponsored by ANRF.

Event Organized – Faculty Development Program



DEPARTMENT OF PHYSICS
Organizes
Seven Day
International Faculty Development Program (FDP) on
**"FRONTIERS IN PHYSICS FROM
FUNDAMENTAL TO EMERGING TECHNOLOGIES"**
(Online Mode)
12-18 May 2025 Time : 6.00 pm - 7.00 pm

Academy of Maritime Education and Training (AMET)
Deemed to be University
#135, East Coast Road, Kanathur, Chennai - 603 112, Tamil Nadu, India
Website : www.ametuniv.ac.in

Invited Speakers Time : 6.00 pm - 7.00 pm

 Dr. T. Senthilnathan Assistant Professor, Department of Applied Physics, Sri Venkateswara College of Engineering, Siparumbudur-602117	Date : 12-05-25
 Dr. Kiran Kumar Tadi Department of Chemistry School of Advanced Sciences Vellore Institute of Technology, Chennai-600127, Tamilnadu, India	Date : 13-05-25
 Dr. Arindam Head, Sensor Development Section Materials Science Group Indira Gandhi Centre for Atomic Research Kalpakam-603102, Tamilnadu, India	Date : 14-05-25
 Dr. S. Anandh Jesuraj Assistant Professor – Research Centre for Nanoscience and Nanotechnology, Sathyabama Institute of Science and Technology, Chennai – 600119 Tamil Nadu, India	Date : 15-05-25
 Dr. Yuvaraj Assistant Professor, Department of Physics Vel Tech Rangarajan Dr. Sagunthala R & D Institute of Science and Technology, Chennai – 600062, Tamil Nadu, India	Date : 16-05-25
 Dr. Tejaswi Ashok Hegde Assistant Professor (SG) Department of Physics Hindustan Institute of Technology and Science Chennai – 603103, Tamil Nadu, India	Date : 17-05-25
 Dr. S. Santhoshkumar Trace Bioanalytical Laboratory Department of Chemistry National Sun Yat Sen University Kaohsiung Taiwan.	Date : 18-05-25

Organizing Committee

Chief Patron :
Dr. J. Ramachandran, Chancellor, AMET
Dr. Rajesh Ramachandran, President, AMET
Mrs. Sushela Ramachandran, Managing Trustee, AMET
Col. Dr. G. Thiruvanasagam, Provost, AMET

Patron :
Prof. Dr. V. Rajendran, Vice Chancellor, AMET

Steering Committee :
Dr. Deepa Rajesh, Vice President (Academics), AMET
Dr. R. Muthuchidamban, Registrar (IG), AMET
Dr. V. Sangeetha Albin, Additional Registrar, AMET
Dr. M. Jayaprakashvel, Special Officer & Coordinator - IQAC, AMET

Convenors :
Dr. A. Mohamed Haroon Basha, Associate Professor, Department of Physics, AMET
Dr. C. Amirthakumar, Assistant Professor, Department of Physics, AMET

Faculty Co-ordinators
Dr. K. Anandhan, Head & Associate Professor, Department of Physics, AMET
Dr. Anila R. Wandas, Professor, Department of Physics, AMET
Dr. A. Mohamed Haroon Basha, Associate Professor, Department of Physics, AMET
Dr. K. Gayathri, Deputy Head & Associate Professor, Department of Physics, AMET
Dr. K. Rajesh, Associate Professor, Department of Physics, AMET
Dr. G. Jayalakshmi, Associate Professor, Department of Physics, AMET
Dr. C. Amirthakumar, Assistant Professor, Department of Physics, AMET
Dr. Sangeeta Mary Gali, Assistant Professor, Department of Physics, AMET
Dr. T.S. Jeevitha, Assistant Professor, Department of Physics, AMET.

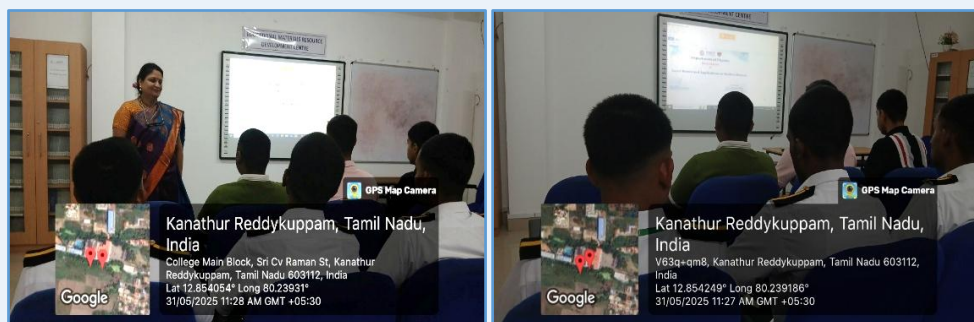
All the correspondence should be addressed to
Dr. A. Mohamed Haroon Basha, Convenor & Associate Professor,
Dr. C. Amirthakumar, Convenor & Assistant Professor, Department of Physics,
AMET, Chennai – 603 112.
Mobile : +91-9248453242, +91-8678925122
E-mail : haroonbasha09@amethuniv.ac.in; amirthakumar@amethuniv.ac.in

ADMISSIONS FOR THE YEAR 2025-2026 ARE OPEN
Toll Free Number : 1800 108 3030
Contact for Admission : +91-93443 91418 (Whatsapp)
Email : admission@amethuniv.ac.in
Facebook Link : <https://www.facebook.com/AMETInternational/>
Instagram : <https://www.instagram.com/amethuniversity/>
Youtube : https://www.youtube.com/channel/UC6Jzt1MSE_0B0uJmMPLzPQ/

Dr. A. Mohamed Haroon Basha and Dr. C. Amirthakumar organized a Seven-Day International Faculty Development Program (FDP) on “Frontiers in Physics: From Fundamentals to Emerging Technologies” from 12th to 18th May 2025. The program witnessed active participation from faculty members across various institutions throughout India.

Event Organized – Guest Lecture

Our Physics Department actively organizes a variety of academic events such as Faculty Development Programs (FDPs), Guest Lectures, Research Methodology Workshops, and other scholarly activities. These initiatives aim to foster a vibrant academic environment by facilitating knowledge exchange, enhancing research skills, and promoting continuous professional development. By conducting these events, the department not only strengthens the academic community but also encourages collaboration, innovation, and excellence among faculty members and students.



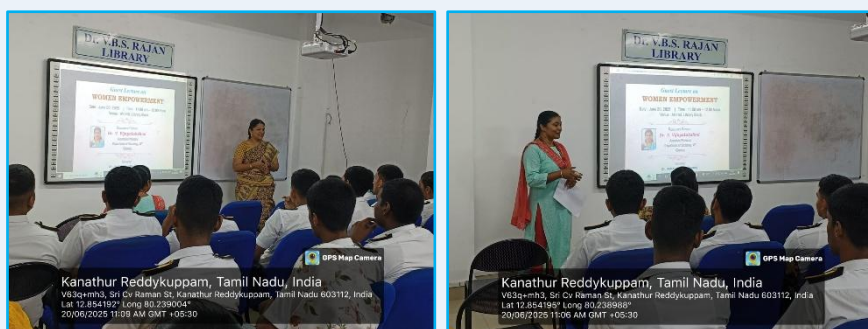
Guest Lecture on "Laser basics and applications in modern science"

A guest lecture on “*Laser Basics and Applications in Modern Science*” was delivered by **Dr. S. Sivagamasundari** for **Nautical Science students** on **31st May 2025**. The event was **organized by Dr. K. Anandan**, providing students with valuable insights into laser technology and its modern applications.



Guest Lecture on "Carbon Nanotube and its importance"

A guest lecture on “*Carbon Nanotube and its importance*” was delivered by **Dr. M. Arivandandam**, Professor, Centre for Nanoscience and Nanotechnology, Anna University for **Nautical Science students** on **21March 2025**. The event was **organized by Dr. K. Rajesh**, providing students with valuable insights into CNT and its modern applications.



Guest Lecture on "Women Empowerment"

A guest lecture on “*Women Empowerment*” was delivered by **Dr. V.Vijayalakshmi** for **Nautical Science students** on **20.06.2025**. The event was **organized by Dr. Anitha Rexalin Devaraj**, providing students with valuable insights women empowerment and gender sensitivity.



Guest Lecture on "Recent Trends in Applied Physics"

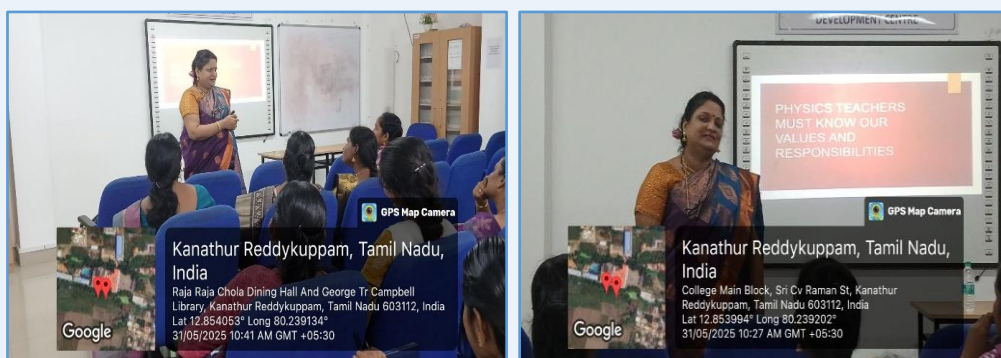
A guest lecture on “**Recent Trends in Applied Physics**” was delivered by **Dr. Tejaswi Ashok Hegde**, Assistant Professor (SG) of Physics, Hindustan University, on **21st May 2025**. The event was convened by **Dr. T. U. Jeevitha**, Assistant Professor. The session provided participants with valuable insights into emerging developments and applications in the field of applied physics.

Event Organized – Professional Ethics Program



1. Professional Ethics Program

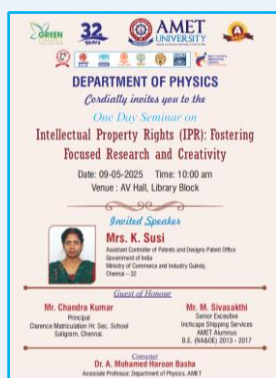
A program on “*Faculty Ethics: Principles, Practice and Professionalism*” was conducted on **3rd May 2025**, featuring **Dr. S. Gunasekaran**, TANSA Awardee and Dean (R&D), St. Peter’s Institute of Higher Education and Research, Chennai, as the guest speaker. The event was **organized by Dr. K. Anandan**, emphasizing the importance of ethics and professionalism in academia.



2. Professional Ethics Program

A session on “*Empowering Faculty through Ethical Practices*” was conducted on **31st May 2025**, with **Dr. S. Sivagamasundari** as the guest speaker. The program was **organized by Dr. K. Anandan**, focusing on the role of ethics in enhancing faculty effectiveness and integrity.

Event Organized – Seminar

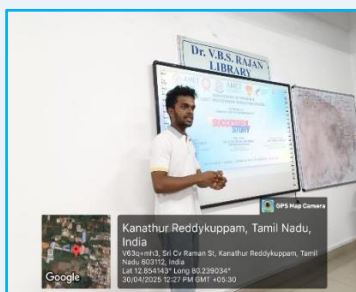


A **One-Day IPR Seminar** on “*Intellectual Property Rights (IPR): Fostering Focused Research and Creativity*” was conducted on **9th May 2025**, organized by **Dr. A. Mohamed Haroon Basha**, Associate Professor.



Seminar on "Upholding Integrity in Research: Navigating Ethics and Combating Plagiarism"

A **Research Seminar** on “*Upholding Integrity in Research: Navigating Ethics and Combating Plagiarism*” on **26/03/2025**. The event was coordinated by **Dr. K. Rajesh**, Associate Professor, Department of Physics, with the objective of promoting awareness and understanding of research ethics and the importance of maintaining academic integrity.



Seminar on Entrepreneurship: My successful Story

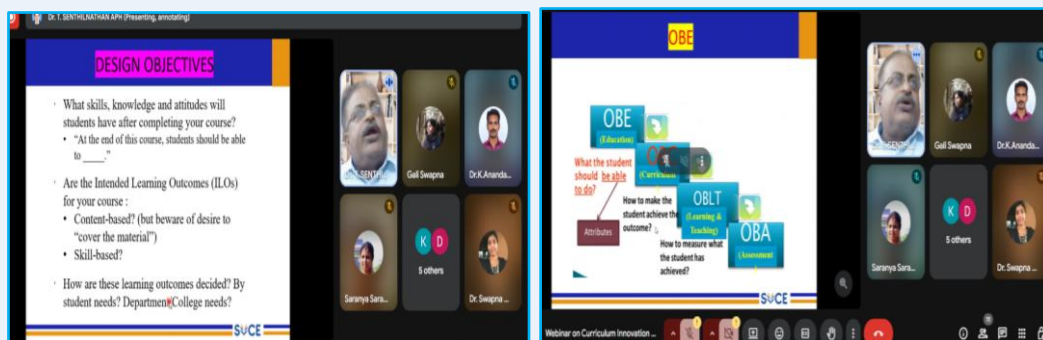
A “*Seminar on Entrepreneurship: My successful Story*” was delivered by **Cdt. Ronal Antony** for **Nautical Science** students on **30.04.2025**. The event was **organized by Dr. Anitha Rexalin Devaraj**, providing students with valuable insights into enter into the field of innovation and entrepreneurship.



Seminar on "Skill Development"

A seminar on “Hands-on Training on LaTeX for Faculty and Researchers” was conducted on **21st May 2025** as part of a seminar on skill development. The session was delivered by **Dr. Tejaswi Ashok Hegde**, Assistant Professor (SG), Hindustan University. The event was convened by **Dr. T. U. Jeevitha**, Assistant Professor, providing faculty and researchers with practical exposure to LaTeX and its applications in academic writing.

Event Organized – Webinar



Webinar on “Curriculum Innovation for Academic Excellence”

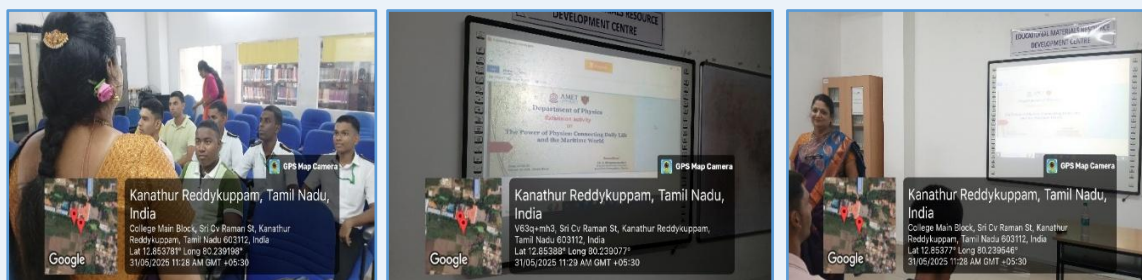
Webinar on “*Curriculum Innovation for Academic Excellence*” was delivered by **Prof. Sentilnathan, S. V.** College of Engineering on **11th June 2025**. The event was **organized by Dr. Swapna Mary Gali**. It highlighted practical strategies like simulation, modular labs, and structured assessment to enhance academic excellence.

Event Organized – Department Symposium



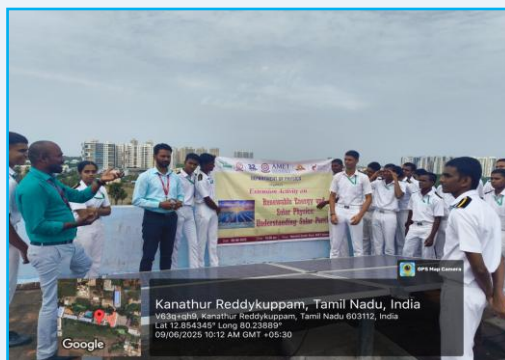
A **symposium on “Physics Empowering Future Technologies”** was organized by **Dr. Swapna Mary Gali, Department of Physics**, on **June 10, 2025**, to promote awareness about the vital role physics plays in our everyday experiences. The event witnessed enthusiastic participation from nautical students and faculty.

Event Organized – Extension Program



Extension Program for Nautical Science Cadets

An extension program titled “*The Power of Physics: Connecting Daily Life and the Maritime World*” was conducted on **31st May 2025** for **Nautical Science students**. **Dr. S. Sivagamasundari** served as the guest speaker, focusing on bridging classroom learning with real-world maritime applications and fostering a scientific mindset. The event was **organized by Dr. K. Anandan**.



Extension Program for Nautical Science Cadets

An extension program titled “**Renewable Energy and Solar Physics: Understanding Solar Panels**” was conducted on **9th June 2025** for **Nautical Science students**. **Dr. K. Rajesh**, Associate Professor and the coordinator of the program, gave an in-depth technical explanation on the working principle of solar panels, including the photovoltaic effect.



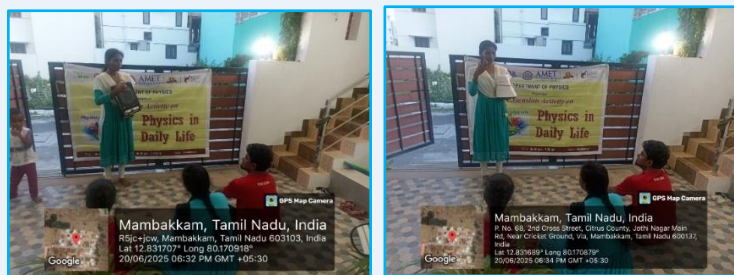
Extension Program on Vitals at a Glance: Free Health Check up

The Department of Physics, successfully organized an Outreach Health Screening Camp titled “**Vitals at a Glance: Free BP, Sugar & ECG Check**” on **20th May 2025**, in collaboration with Gleneagles Health City, Perumbakkam, Chennai. **Dr.K.Gayathri**, the initiative aimed to promote preventive healthcare awareness among the local Kanathur community and AMET employees by offering essential health screenings at no cost. **Over 150 individuals benefited from the camp**, which saw active participation from students, faculty, administrative staff, and members of the local community.



Extension Program for IIT & IITDM

An extension program titled “*Incubation visit to ACEII*” was arranged on **24.06.2025** for **IIT, Chennai and IITDM, Kancheepuram students**. It focusing on bridging classroom learning with real-world maritime applications and fostering a scientific mindset in the field of Entrepreneurship. The event was **organized by Dr. Anitha Rexalin Devaraj**.



An extension program titled “**Physics in Daily Life**” was conducted by **Dr.Swapna Mary Gali** on **20th June 2025** for public. This outreach initiative bridged theoretical physics with tangible applications, making learning both accessible and relevant for students and community participants.



Extension Program on Pendulum Physics in the Park

An extension activity titled “**Pendulum Playground – Fun with Physics in the Park**” was conducted on **10th June 2025**. The event aimed to engage participants with interactive demonstrations of pendulum motion in an open park setting, making physics fun and accessible. It was organized by **Dr. T. U. Jeevitha**, Assistant Professor, fostering scientific curiosity through hands-on learning beyond the classroom.

Event Organized – Outreach Program



Outreach Program for Marine Engineering Cadets

An Outreach program titled “**Know your Food: A day at the Poultry Form**” was conducted on **9th June 2025** for **Nautical Science students** at JK Poultry form, Uthandi. The event was coordinated by Dr. K. Rajesh, Associate Professor, Department of Physics, and saw the enthusiastic participation of 40 students from the Marine Engineering Department.



Outreach Program on Applied Physics in Healthcare

The Department of Physics organized an insightful extension program titled “**Applied Physics in Healthcare: An Extension Initiative**” in academic partnership with Gleneagles HealthCity, Perumbakkam, on **20th May 2025**. Convened by **Dr.K.Gayathri**, Deputy Head & Associate Professor, the program featured sessions on the basic principles of radiation physics and image formation in X-Ray & CT scans, along with electrical conductivity in the human body and waveform interpretation in ECG.



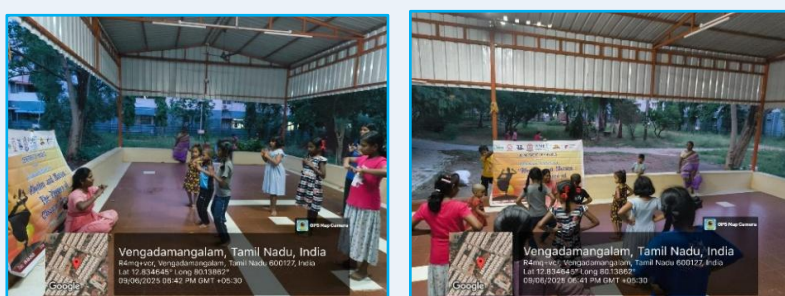
Incubation Centre visit – Anna University

An Outreach program titled “*Incubation Visit to Anna University Incubation center*” was organized on **27.05.2025** for **various department students** those who are interested to startup companies was **organized by Dr. Anitha Rexalin Devaraj**.



Yoga Asanas for stress relief and wellness

An outreach program titled “**Yoga asanas for stress relief and wellness**” was conducted by **Dr. Swapna Mary Gali** on **21st June 2025** for public The session emphasized simple yet effective techniques to combat stress and improve overall well-being, especially relevant in the context of modern academic and professional pressures.



Outreach Program on Exploring the Physics of Classical Dance

An outreach activity titled “**Rhythm and Motion: The Physics of Classical Dance**” was conducted on **9th June 2025**. The session explored the intersection of classical dance and physics, highlighting the scientific principles behind rhythm, balance, and movement. The event was **organized by Dr. T. U. Jeevitha**, as part of efforts to promote interdisciplinary learning and appreciation of physics in the arts.

